Risk Management: Assignment 2

Financial Engineering

Corporate Bonds & Single Factor Model

Instructions

- **Delivery:** Friday 18:00 p.m. the 27th of March to financial.engineering.polimi@gmail.com, with subject "RM: Assignment 2, Group #";
- Deliver
  a. a **short** pdf document with the results. List the errors you've found.
  b. the code written in Python using your preferred IDE;
- Fill the gaps where appropriate and correct the errors (in case of incoherences between this document and the code, follow this document);
- Comment (in detail) the code and use explicative variable names;
- Use the data and the curve bootstrapped during the Risk Management: Assignment 0.

Case study A

On the 15th of February 2008 the macro hedge fund Polimi Capital has the following position on fixed rate corporate bonds issued by the firm Beta

- Long €10m (face value) of a bond with 1 year maturity, 5.5% coupon with semi annual (s/a) payments, market dirty price of €100.00;
- Long €10m of a bond with 2 years maturity, 7% coupon with s/a payments, market dirty price of €103.00.

Questions

1. Derive the Z-spreads for each bond;
2. Derive two distinct intensity values for the two bonds under the assumption that intensity is constant in time and with a recovery rate of 40%. Derive the default probability of Beta at one and two years. Do you expect the two intensities to be equal?
3. Assume now that the default intensity of Beta is not constant and derive the default probability of Beta at one and two years from a piece-wise constant intensity. Which bootstrap approach would give a better approximation of Beta's creditworthiness?
4. Consider scenario 1 in which the market expectation of the medium-term survival probability of Beta changes instantaneously, so that the dirty price of the two-years bond drops to 97 (price of the one-year bond unchanged). Execute a new bootstrap of the intensity and derive a new estimate of the default probabilities at 1 and 2 year;
5. Derive the historical default probability of Beta at one and two years and the default probability between the first and second year (contitional to the survival in the first year) from the rating transition matrix at one year that you find in the code and compare to the conditional probability derived under scenario 1.

6. Consider now a portfolio made up of 100 zero-coupon bonds with 2y maturity that pay €1Mln at maturity, issued by different corporations with initial rating Investment Grade. The defaults and downgrades of the issuers are modelled with a single factor firm model:

$$v_i = \rho_i y + \sqrt{1 - \rho_i^2}\epsilon_i;$$

- Presence of default and migration risk, described by the transition matrix (IG, HY, Def);
- Recovery rate π = 40% for all issuers

a. Mark-to-market the portfolio (all issuers are IG);
b. Derive the 1y **forward** price of a zero-coupon bond with one year residual maturity, in the cases that the issuer one year from today is (i) investment grade and (ii) high yield;
c. Execute Monte-Carlo simulations with (i) ρ = 0%, (ii) ρ = 25% (iii) ρ = 50% using 1,000,000 scenarios. Count the average number of defaults and downgrade events, evaluate 99% VaR considering (i) only defaults, (ii) defaults and migrations.
d. Rank the VaRs in and discuss the following topics: the relevance of migration risk vs. default risk and the relevance of the level of correlation.